

Excom Calls — Implementation Spec & MVP Blueprint

Provider-Agnostic Voice Telephony Channel Integrated into Unified Excom Timeline

COMPONENT	PREPARED FOR	DEVELOPER	ARCHITECTURE VERSION
excom / channels / voice	Sagar · GGIL	Somil Vaishya	v1.0 (Dual Provider Ready)

01 · Governing Architectural Principles

1. The "Dumb-Pipe" Rule

Excom owns 100% of business decisions: who rings, routing hierarchy, queue priorities, recordings, and timeline storage. Telecom providers (Exotel, Airtel IQ) are strictly treated as dumb pipes that execute call instructions.

2. The "Two-Places" Rule (Zero Vendor Lock-in)

The vendor string `"Exotel"` is restricted to appear in exactly **TWO** locations across the entire codebase:

- 1. `channels/voice/providers/exotel.py` (The adapter implementation)
- 2. `voice_provider` Select field on `Excom Channel Account` (System Manager view only)

All user-facing UI labels, API endpoints, doctypes, and events say **"Calls"**. Switching to Airtel IQ requires writing **1 adapter file** and selecting the new dropdown option — zero database migrations.

02 · Architecture & Code Structure

Voice sits directly inside `excom` alongside `email` and `whatsapp` without creating separate apps or extra bench installations:

```
excom/excom/
├── channels/
│   └── voice/
│       ├── __init__.py
│       ├── handler.py      # Inbound webhook receiver (Token & Guest auth)
│       ├── outbound.py    # Click-to-call dialer, bridge initiator & hangup
│       ├── routing.py     # Decision engine (Sticky-then-team <5s cache-only)
│       ├── recording.py   # Authenticated recording streaming proxy
│       └── providers/
│           ├── base.py    # VoiceProvider ABC & CallEvent dataclasses
│           ├── exotel.py  # Exotel Connect Applet Adapter
│           └── airtel.py  # Airtel IQ Adapter (Stub for future switch)
└── api/
    └── voice.py           # Whitelisted client APIs for desk UI & active widgets
```

03 · Data Model (Single New DocType: Excom Call)

Why only 1 DocType? Agents are core Frappe `User` records, Routing lives on `Excom Channel` `Account.allowed_teams`, and call records need dedicated indexed columns for analytics (duration, cost, talk-time, ring-time) that mutate 4–6 times per call.

Field Name	Field Type	Purpose & Behavior
<code>name</code>	Hash	Independent internal ID (never tied to vendor ID to prevent vendor lock).
<code>provider_call_id</code>	Data (Unique Index)	The vendor's <code>CallSid</code> . Serves as the idempotency key for safe webhook retries.
<code>direction</code>	Select	<code>Inbound</code> <code>Outbound</code>
<code>from_number</code> / <code>to_number</code>	Data	Standardized E.164 phone numbers (e.g. +919876543210).
<code>business_number</code>	Data	The ExoPhone DID number that received or placed the call. Resolves account & team.
<code>agent</code> / <code>team</code>	Link	Linked to <code>User</code> and <code>Excom Team</code> .
<code>status</code>	Select	<code>Ringing</code> → <code>In-progress</code> → <code>Completed</code> → <code>Missed</code> → <code>Failed</code>
<code>duration</code> / <code>talk_time</code>	Int (Seconds)	Backfilled accurately ~2 minutes post-call by the background reconcile cron job.
<code>recording_url</code>	Data	Internal storage reference. Never rendered raw to browser; streamed via auth proxy.
<code>thread</code> / <code>identity</code>	Link	Ties the call into the Unified Timeline alongside WhatsApp and Email threads.

04 · Inbound & Outbound Call Mechanics

A. Inbound Routing: "Sticky-then-Team, First-Answer-Wins"

1. **Customer Dials DID** → Provider hits Exotel Connect Dynamic URL GET endpoint.
2. **Excom Routing Engine (< 5s Budget):**
 - Checks if caller spoke to an active agent recently (**Sticky Agent**). If yes, rings that agent first.
 - If no previous agent or no answer, triggers **Parallel Ringing** across all team members in `allowed_teams`.
 - **Strict Constraint:** The endpoint performs **ZERO inline DB writes**; it queries memory/cache and immediately returns the JSON destination array.
3. **First-Answer-Wins:** The first agent to pick up gets bridged. Realtime broadcast notifies other agents: "Call answered by Agent X".

B. Outbound Calling: "Click-to-Call (PSTN Bridge)"

1. Agent clicks the **Call Button** on any Contact, Lead, or Omni Identity in Excom.
2. Excom invokes Provider API: `From = agent_mobile`, `To = customer_mobile`, `CallerId = ExoPhone_DID`.
3. Agent's mobile phone rings first. As soon as the agent answers, the provider bridges the call to the customer.
4. Provides crystal clear carrier-grade audio quality without requiring WebRTC browser microphone permissions or extra VoIP licenses.

05 · Security, Recording & Governance

Control Area	Specification & Implementation
Authenticated Recording Proxy	Recordings require provider Basic Auth. Excom streams audio via <code>/api/method/excom.excom.api.voice.get_recording</code> after verifying user permission. Raw URLs are never exposed.
3-Level Recording Policy	Configurable at: Account Level (All/Never/Inbound/Outbound) → Route Level (override for specific lines) → Agent Level .
International Spend Caps	3-layer enforcement: Per-agent permission → Account ceiling → Daily minute spend cap (hard block upon limit to prevent 5-figure billing leaks).
Webhook Security	Token verification via query parameter + Provider Egress IP Allowlist + Integration Request logging.

06 · Phase-wise MVP Delivery Blueprint

PHASE 1: Core Backend Infrastructure (Cycle 1)

🕒 Estimated Duration: 4–5 Days

- **✓ DocType Creation:** Build `Excom Call` with all indexes, state machine, and child event logs.
- **✓ Channel Account Extension:** Add `voice_section` in `Excom Channel Account` with credential ping verification.
- **✓ VoiceProvider Interface:** Implement `VoiceProvider` ABC (6 standard methods) + `ExotelAdapter` in `providers/exotel.py`.
- **✓ Inbound Routing Engine:** High-speed, cache-only `routing.py` returning Exotel Connect Dynamic URL JSON payload under 5 seconds.
- **✓ Outbound Dialing Service:** `outbound.py` supporting PSTN 2-leg bridge initiation via API.
- **✓ Async Reconcile Background Job:** Scheduled cron job polling Call Details API to backfill talk time, duration, and cost.
- **✓ Recording Auth Stream Proxy:** Secure streaming endpoint in `recording.py` with download/play permission splits.
- **✓ Timeline Linkage:** Automatic creation of `Excom Message` with `message_type: "Call"` pointing to the Call record.

Phase 1 Exit Criteria: An inbound call successfully rings the team, connects to the agent, registers on the thread, and updates exact duration & cost post-reconcile.

PHASE 2: Frontend Desk UI & Operational Controls (Cycle 2)

🕒 Estimated Duration: 4–5 Days

- **✓ Click-to-Call Surfaces:** Native Call buttons on Thread header, Contact cards, Lead views, and dialpad modal.
- **✓ Active-Call Floating Widget:** Draggable, persistent bottom widget displaying caller ID, linked CRM document, live timer, and Hangup button.
- **✓ Realtime Screen Pop:** Instant desktop popup via `frappe.publish_realtime` fired before DB write when an incoming call rings.
- **✓ Unified Timeline Call Card:** Custom rendered Call bubble in thread with direction badge, duration, and inline waveform audio player.
- **✓ Missed Call Queue:** Dedicated worklist for missed calls with 1-click callback and assignee tracking.
- **✓ International Spend & Policy Controls:** Server-side validation against user calling profiles and daily international caps.

Phase 2 Exit Criteria: Complete end-to-end user experience inside Excom Desk with click-to-call, screen pop, call recording playback, and missed call management.

PHASE 3: Advanced & Future Enhancements (Post-MVP / Cycle 3)

📅 On-Demand / Post-Launch

- ♦ **WebRTC Softphone:** In-browser VoIP calling (requires Exotel SDK agreement & browser SIP credentials).
- ♦ **Excom IVR Visual Builder:** Custom IVR tree builder (Greeting → Keypad Options → Team Routing).
- ♦ **Airtel IQ Provider Adapter:** Implement `providers/airtel.py` for multi-carrier redundancy.
- ♦ **AI Transcription & Sentiment Analysis:** Automatic call speech-to-text transcription via Whisper / Gemini.

07 · Prerequisites Checklist (What is Required from Business)

Prerequisite Item	Source / Owner	Impact / Why it is Required
Exotel KYC Approval	Telecom Provider (Exotel)	BLOCKING Outbound PSTN calling is legally blocked by telecom operators until KYC is cleared.
Exotel API Credentials	Exotel Dashboard	Account SID, API Key, API Secret & Cluster Base URL (e.g. <code>api.in.exotel.com</code>).
ExoPhone DID Numbers	Exotel Dashboard	Virtual business phone numbers assigned to Excom Channel Accounts.
<code>parallel_ringing</code> Activation	Exotel Account Manager	Required for simultaneous team ringing (first-answer-wins). Must be enabled by account manager.
Agent Mobile Numbers	Frappe User DocType	All agents in <code>allowed_teams</code> must have valid mobile numbers in E.164 format (+91...) on their User record.
Public Webhook URL	Hosting / Frappe Cloud	Public HTTPS domain to receive Exotel Connect Dynamic URL hits and StatusCallbacks.